

Threat Hunting and Incident Reconstruction with Splunk BOTS v3

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Executive Summary

Security Information and Event Management (SIEM) platforms have become essential components of modern cybersecurity operations by enabling organisations to collect, centralise, analyse and correlate security events generated across different systems.

This laboratory focused on the practical use of Splunk Enterprise for threat hunting and incident investigation using the Boss of the SOC (BOTS) Version 3 dataset.

The investigation began with the deployment and configuration of Splunk Enterprise on an Ubuntu Server virtual machine. Following successful installation, the BOTS Version 3 dataset was imported and validated to ensure that historical security events were available for analysis. Exploratory data discovery was then performed to identify available sourcetypes, host coverage, important fields and overall event distribution within the dataset.

A hypothesis-driven investigation was subsequently conducted to determine whether an external actor had performed reconnaissance against the organisation's public web application. Splunk Search Processing Language (SPL) was used to analyse web-related logs, identify suspicious request patterns and correlate evidence across multiple security telemetry sources. The investigation was extended through cross-source incident reconstruction to establish an evidence-based timeline describing the observed activity.

Finally, a Splunk investigation dashboard and a historical detection prototype were developed to demonstrate how similar activities could be monitored within a production Security Operations Centre (SOC). Although the investigation utilised historical laboratory data, it demonstrated the effectiveness of Splunk Enterprise for log correlation, behavioural analysis and digital forensic investigations.

Learning Objectives

- Install and verify a pre-indexed Splunk dataset without re-ingesting it.
- Use SPL for field discovery, parsing, aggregation, time analysis and cross-source pivots.
- Develop and test a threat-hunting hypothesis.
- Reconstruct an evidence-backed incident timeline across multiple sourcetypes.
- Create a dashboard and validate an alert against historical data.
- Explain search limitations, false-positive risks and productionisation requirements.

Tools Used

- Oracle VirtualBox: Hosted the Ubuntu Server virtual machine
- Ubuntu Server: Operating system hosting Splunk Enterprise
- Splunk Enterprise: Security Information and Event Management (SIEM) platform
- BOTS Version 3 Dataset: Security event dataset used throughout the investigation

Methodology

The first stage of this laboratory involved the deployment of Splunk Enterprise on an Ubuntu Server virtual machine hosted within Oracle VirtualBox. Before installation, the virtual machine networking was configured using both a Host-Only Adapter and a Network Address Translation (NAT) Adapter. The Host-Only Adapter enabled communication between the Windows operating system and the Ubuntu Server, while the NAT Adapter provided trusted Internet connectivity required for downloading Splunk Enterprise and other supporting resources.

Following the successful configuration of the network interfaces, Splunk Enterprise Version 10.4.1 was downloaded directly from the official Splunk website using the Linux command-line interface. The compressed installation package was extracted into the /opt directory before the Splunk service was started for the first time.

Wget

```
https://download.splunk.com/products/splunk/releases/10.4.1/linux/splunk-10.4.1-5a009d941268-linux-amd64.tgz
sudo tar -xzf splunk-10.4.1-5a009d941268-linux-amd64.tgz -C /opt
sudo /opt/splunk/bin/splunk start --accept-license
```

Following installation, Splunk Enterprise was launched and accessed through the web interface using the default management port 8000. The web interface was used throughout the laboratory to execute SPL searches, investigate historical security events and develop dashboards.

Successful access to the Splunk dashboard confirmed that the installation process had completed successfully and that the SIEM platform was operational.

After installation, the version of Splunk Enterprise was verified to ensure that the correct software release had been installed successfully.

```
splunk releases are running in 100% 100% 100% 100% 100% 100% 100% 100%
socadmin@Splunk-SIEM:~$ sudo /opt/splunk/bin/splunk version
Splunk 10.4.1 (build 5a009d941268)
socadmin@Splunk-SIEM:~$
```

/opt/splunk/bin/splunk version

The output confirmed that Splunk Enterprise Version 10.4.1 had been installed successfully. The operational status of the Splunk service was subsequently verified before accessing the web interface.

```
prompt] [-K directory] [-T timeout] [-u user] file ...
socadmin@Splunk-SIEM:~$ sudo /opt/splunk/bin/splunk status
splunkd is running (PID: 1004).
splunk helpers are running (PIDs: 1005 1300 1302 1399 1465 1572).
socadmin@Splunk-SIEM:~$
```

Confirmation that splunk its running on the Ubuntu server *sudo /opt/splunk/bin/splunk status*

Download, Verify and Install BOTS v3

The BOTS Version 3 dataset was then imported into Splunk Enterprise. Before extraction, the Splunk service was stopped to prevent file conflicts while the dataset was copied into the Splunk application directory. Once the extraction process had completed, file ownership permissions were updated before restarting the Splunk service.

```
sudo /opt/splunk/bin/splunk stop
sudo tar -xzf botsv3_data_set.tgz -C /opt/splunk/etc/apps
sudo chown -R splunk:splunk /opt/splunk/etc/apps
sudo /opt/splunk/bin/splunk start
```

```
RTC in local TZ: no
socadmin@Splunk-SIEM:~$ md5sum botsv3_data_set.tgz
d7ccca99a01cff070dff3c139cdc10eb  botsv3_data_set.tgz
socadmin@Splunk-SIEM:~$ _
```

md5sum botsv3_data_set.tgz: confirmation of the integrity of the downloaded file

To validate that the dataset had been indexed correctly, an initial search was executed against the botsv3 index.

The screenshot shows the Splunk Enterprise search interface. The search bar contains the following query:

```
index=botsv3 earliest=0  
| stats count min(_time) as earliest max(_time) as latest dc(sourcetype) as  
sourcetypes  
| convert ctime(earliest) ctime(latest)
```

The search results show 1,944,092 events. The results table is displayed with the following columns: count, earliest, latest, and sourcetypes. The first row shows a count of 1944092, with the earliest time being 08/20/2018 04:00:03 and the latest time being 09/19/2019 18:10:50. The total number of sourcetypes is 107.

count	earliest	latest	sourcetypes
1944092	08/20/2018 04:00:03	09/19/2019 18:10:50	107

```
index=botsv3 earliest=0  
| stats count min(_time) as earliest max(_time) as latest dc(sourcetype) as sourcetypes  
| convert ctime(earliest) ctime(latest)
```

Data Discovery and Source Inventory

Following the successful deployment of Splunk Enterprise and the validation of the imported BOTS Version 3 dataset, an exploratory analysis was performed to understand the available security telemetry before commencing the threat hunting investigation. This phase of the investigation focused on identifying the available log sources, determining host coverage, discovering commonly populated fields and analysing the overall distribution of security events within the dataset. Understanding the available evidence before conducting any investigation is considered an important step in digital forensics because it enables investigators to formulate realistic hypotheses based on actual data rather than assumptions.

Sourcetype inventory

The analysis commenced with an inventory of the available sourcetypes within the botsv3 index. Sourcetypes represent the different categories of logs collected by Splunk and provide an overview of the available evidence that can support an investigation. By identifying the available sourcetypes together with their event counts and time coverage, it became possible to determine which data sources were most suitable for subsequent threat hunting activities.

```
index=botsv3 earliest=0
```

```
| stats count as events earliest(_time) as first latest(_time) as last by sourcetype
```

```
| convert ctime(first) ctime(last)
```

```
| sort - events
```

sourcetype	events	first	last
syslog	283976	08/20/2018 09:04:14.000000000	08/20/2018 15:17:59.000000000
stream:ip	227872	08/20/2018 09:00:20.177	08/20/2018 15:27:09.296000000
osquery:results	219997	08/20/2018 04:00:03.000000	08/20/2018 15:17:55.000000
stream:dns	218456	08/20/2018 09:03:42.964	08/20/2018 15:18:00.966
stream:udp	157960	08/20/2018 09:03:46.095	08/20/2018 15:27:09.296000000
PerfmonMk:Process	139828	08/20/2018 09:04:21	08/20/2018 15:17:59
WinHostMon	129679	08/20/2018 09:08:23.000	08/20/2018 15:17:27.000
aws:cloudwatchlogs	115145	08/20/2018 09:23:28.000	08/20/2018 15:08:14.000
aws:cloudwatchlogs:vpflow	97448	08/20/2018 09:00:13.000	08/20/2018 15:26:46.000
stream:tcp	84031	08/20/2018 09:01:10.111999000	08/20/2018 15:18:00.000000000
osquery:info	83961	08/20/2018 09:04:40.772761	08/20/2018 15:17:55.951526
cisco:asa	80192	08/20/2018 09:04:13.000000	08/20/2018 15:18:00.000000

Sourcetype Inventory Generated from the BOTS Version 3 Dataset

Host coverage

Following the identification of available sourcetypes, the next stage involved determining the distribution of events across the various hosts contained within the dataset. Host coverage analysis enables investigators to understand which systems generated security events and identifies systems that may require further investigation due to unusually high event volumes or limited logging activity.

```
index=botsv3 earliest=0
```

```
| eval host=coalesce(host, ComputerName, dvc, dest)
```

```
| stats count as events dc(sourcetype) as source_types values(sourcetype) as
```

```
| sort - events
```

The screenshot shows the Splunk Enterprise search interface. The search bar contains the following query:

```
index=botsv3 earliest=0
| eval host=coalesce(host, ComputerName, dvc, dest)
| stats count as events dc(sourcetype) as source_types values(sourcetype) as
sourcetypes by host
| sort - events
```

The search results show 2,083,056 events. The results are displayed in a table with the following columns: host, events, source_types, and sourcetypes. The table shows results for the host 'hoth'.

host	events	source_types	sourcetypes
hoth	382718	40	Unix:ListeningPorts Unix:SSHDCConfig Unix:Service Unix:UserAccounts access_combined

Host Coverage Analysis

Field discovery

To better understand the structure of the imported dataset, a field discovery exercise was subsequently performed using the `fieldsummary` command. Unlike previous searches which focused on events, this search examined the metadata associated with each field contained within the indexed events. Field discovery enables investigators to identify available field names, determine field popularity and discover equivalent fields representing the same information under different names.

```
index=botsv3 earliest=0  
| fieldsummary  
| sort - count
```

The screenshot shows the Splunk Enterprise search interface. The search bar contains the query: `index=botsv3 earliest=0 | fieldsummary | sort - count`. The search results are displayed in a table format, showing statistics for 2,034 events. The table has columns for field, count, distinct_count, is_exact, max, mean, min, and numeric_count. The results are sorted by count in descending order.

field	count	distinct_count	is_exact	max	mean	min	numeric_count
host	1876447	30	1				0
index	1876447	1	1				0
linecount	1876447	260	0	6424	2.6940792892098737	1	1876447

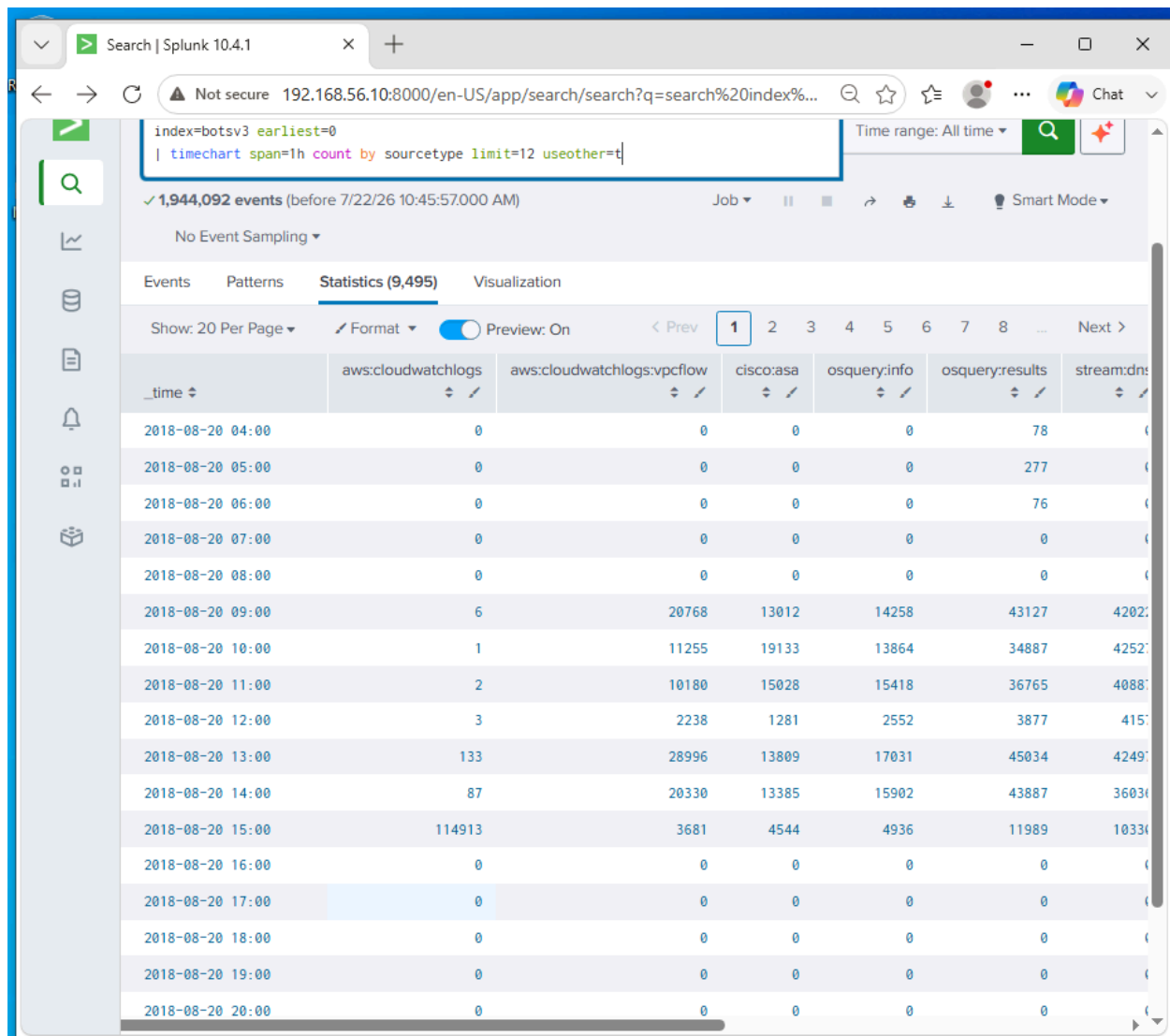
Field Discovery Results

Event volume over time

Having established an understanding of the available fields, the final stage of the exploratory analysis examined the overall distribution of security events over time. Temporal analysis provides investigators with a baseline understanding of normal activity levels and assists in identifying periods characterised by unusual increases or decreases in event volume.

index=botsv3 earliest=0

| timechart span=1h count by sourcetype limit=12 useother=t



Event Volume Distribution Over Time

The exploratory analysis completed in this section established a comprehensive understanding of the available evidence before commencing the hypothesis-driven threat hunting investigation. By identifying available log sources, validating field names, analysing host coverage and examining event distribution over time, a reliable baseline was established for the subsequent investigation presented in the following section of this report.

Hypothesis-Driven Hunt: Suspicious Web Activity

To investigate this hypothesis, attention was directed towards web-related telemetry contained within the BOTS Version 3 dataset. Since different web technologies frequently store similar information using different field names, field normalisation was first performed using the `coalesce()` function. This allowed equivalent fields from different sourcetypes to be combined into common variables representing the source IP address, requested URI and HTTP response status.

```
index=botsv3 earliest=0 (sourcetype=*web* OR sourcetype=*access* OR sourcetype=*stream:http*)
| eval src_ip=coalesce(src, src_ip, c_ip, clientip)
| eval uri_path=coalesce(uri, uri_path, url, cs_uri_stem)
| eval http_status=coalesce(status, sc_status, status_code)
| stats count as requests dc(uri_path) as unique_paths values(http_status) as statuses by src_ip
| where requests > 20
| sort - requests
```

The screenshot displays the Splunk Enterprise web interface. A new search is being executed with the following SPL query:

```
index=botsv3 earliest=0 (sourcetype=*web* OR sourcetype=*access* OR sourcetype=*stream:http*)
| eval src_ip=coalesce(src, src_ip, c_ip, clientip)
| eval uri_path=coalesce(uri, uri_path, url, cs_uri_stem)
| eval http_status=coalesce(status, sc_status, status_code)
| stats count as requests dc(uri_path) as unique_paths values(http_status) as statuses by src_ip
| where requests > 20
| sort - requests
```

The search results are displayed in a table with the following columns: src_ip, requests, unique_paths, and statuses. The results are sorted by the number of requests in descending order.

src_ip	requests	unique_paths	statuses
172.16.0.149	4600	133	200, 304, 404
172.16.1.239	2616	1	200
192.168.3.130	2331	1023	200, 201, 204, 206, 301, 302, 303, 304, 404, 408
172.16.197.137	841	425	200, 204, 206, 301

Initial Threat Hunting Search for Suspicious Web Activity

Discovering Actual Field Names and Inspecting Representative Raw Events

Before beginning the investigation, it was necessary to identify the actual field names used in the BOTS Version 3 dataset rather than assuming that every web-related log source used identical field names. This step is important because different applications and web servers often record similar information under different field names, which can affect the accuracy of subsequent searches.

Following the field discovery exercise, representative raw events were inspected to validate that the identified fields accurately represented the original log entries and to understand the structure of the available web logs.

```

index=botsv3 earliest=0
(sourcetype=*web* OR sourcetype=*access* OR sourcetype=*stream:http*)
|fieldsummary
| head 20

```

The screenshot shows the Splunk Enterprise search interface. The search bar contains the command: `index=botsv3 earliest=0 (sourcetype=*web* OR sourcetype=*access* OR sourcetype=*stream:http*) | fieldsummary | sort - count`. The search results show 31,423 events. The 'Statistics (116)' tab is active, displaying a table with columns: field, count, distinct_count, is_exact, max, mean, min, numeric_count, stdev, and values. The 'Root' field has a count of 2049, distinct_count of 500, and is_exact of 0. The 'values' column shows a list of hexadecimal strings.

field	count	distinct_count	is_exact	max	mean	min	numeric_count	stdev	values
Root	2049	500	0				0		[{"value": "1-51c50bd2e3461555b58c150-682ce"}, {"value": "1-5t6d78f050de2cfc5b58c150-bfc52"}, {"value": "1-5tffff90d0487dct5b58c151-3f241"}, {"value": "1-5t186f1d40d810635b58c156-45da1"}, {"value": "1-5t57c8431aef7015"}]

For easy inspection, this command was used to filter for the important field in the dataset

```

index=botsv3 earliest=0
(sourcetype=*web* OR sourcetype=*access* OR sourcetype=*stream:http*)
| fieldsummary
| search field=src OR field=src_ip OR field=clientip OR field=c_ip OR field=uri OR field=url OR field=uri_path OR field=status
OR field=sc_status

```

index=botsv3 (sourcetype=*web* OR sourcetype=*access* OR sourcetype=*stream:http*)

fieldsummary

search fieldsrc OR ffieldsrc_ip OR fieldclientip OR fieldc_ip OR fielduri OR fielduri1 OR fielduri_path OR fieldc_uri_stem OR fieldstatus OR fieldc_status OR fieldstatus_code

31,423 events (before 7/22/26 11:41:22.000 AM)

No Event Sampling

Job

Smart Mode

Events

Patterns

Statistics (6)

Visualization

Show: 20 Per Page

Format

Preview: On

field	count	distinct_count	is_exact	max	min	numeric_count	stdev	values
c_ip	2474	60	1			0		["value":172.16.0.149,"count":590],["value":172.16.1.239,"count":590],["value":172.31.12.76,"count":163],["value":192.168.8.112,"count":145],["value":192.168.3.130,"count":124],["value":172.16.0.178,"count":90],["value":172.16.197.137,"count":89],["value":192.168.8.183,"count":80],["value":192.168.70.186,"count":76],["value":192.168.8.111,"count":72],["value":172.16.133.131,"count":69],["value":172.16.0.127,"count":55],["value":192.168.24.128,"count":49],["value":172.16.0.13,"count":44],["value":192.168.247.129,"count":42],["value":192.168.247.131,"count":39],["value":192.168.8.109,"count":38],["value":192.168.105.214,"count":28],["value":192.168.105.215,"count":18],["value":172.16.0.145,"count":15],["value":172.16.0.109,"count":14],["value":45.7.231.174,"count":4],["value":118.111.168.230,"count":2],["value":192.168.8.117,"count":2],["value":61.75.35.114,"count":2],["value":183.233.123.104,"count":1],["value":1137.74.30.68,"count":1],["value":138.204.135.210,"count":1],["value":138.255.164.126,"count":1],["value":141.212.122.144,"count":1],["value":153.197.134.1,"count":1],["value":156.202.170.156,"count":1],["value":156.205.129.198,"count":1],["value":156.220.233.101,"count":1],["value":172.31.38.181,"count":1],["value":177.184.188.153,"count":1],["value":182.48.105.210,"count":1],["value":186.188.197.47,"count":1],["value":186.236.28.164,"count":1],["value":189.27.101.5,"count":1],["value":189.60.122.174,"count":1],["value":189.89.84.1,"count":1],["value":192.168.9.30,"count":1],["value":196.52.43.70,"count":1],["value":200.122.200.14,"count":1],["value":201.1.55.176,"count":1],["value":201.150.52.35,"count":1],["value":201.203.230.86,"count":1],["value":201.69.148.104,"count":1],["value":220.145.38.79,"count":1],["value":31.184.195.100,"count":1],["value":36.80.156.172,"count":1],["value":36.82.102.227,"count":1],["value":41.189.191.67,"count":1],["value":46.16.231.197,"count":1],["value":58.211.79.35,"count":1],["value":78.150.44.4,"count":1],["value":78.162.195.151,"count":1],["value":89.242.15.50,"count":1],["value":92.243.55.42,"count":1]]
clientip	3907	22	1			0		["value":172.16.0.149,"count":2818],["value":172.16.1.239,"count":833],["value":192.168.8.183,"count":87],["value":192.168.8.112,"count":79],["value":192.168.8.111,"count":40],["value":192.168.8.109,"count":35],["value":183.233.123.104,"count":1],["value":141.212.122.144,"count":1],["value":153.197.134.1,"count":1],["value":156.202.170.156,"count":1],["value":117.184.188.153,"count":1],["value":166.236.28.164,"count":1],["value":189.27.101.5,"count":1],["value":196.52.43.70,"count":1],["value":201.1.55.176,"count":1],["value":201.150.52.35,"count":1],["value":201.203.230.86,"count":1],["value":201.69.148.104,"count":1],["value":220.145.38.79,"count":1],["value":31.184.195.100,"count":1],["value":36.80.156.172,"count":1],["value":36.82.102.227,"count":1],["value":41.189.191.67,"count":1],["value":46.16.231.197,"count":1],["value":58.211.79.35,"count":1],["value":78.150.44.4,"count":1],["value":78.162.195.151,"count":1],["value":89.242.15.50,"count":1],["value":92.243.55.42,"count":1]]
src_ip	10398	94	1			0		["value":192.168.3.130,"count":2207],["value":172.16.1.239,"count":1193],["value":172.16.0.149,"count":1192],["value":91.207.175.249,"count":839],["value":172.16.197.137,"count":752],["value":192.168.24.128,"count":532],["value":192.168.70.186,"count":430],["value":172.16.133.131,"count":302],["value":172.31.12.76,"count":238],["value":112.196.122.127,"count":237],["value":192.168.247.131,"count":216],["value":135.162.246.227,"count":197],["value":192.168.247.129,"count":193],["value":45.67.40.155,"count":183],["value":103.168.8.117,"count":165],["value":172.16.0.178,"count":155]

Actual field names with fieldsummary and inspection on representative raw events were also carried out

To identify systems generating unusually high volumes of web traffic, HTTP requests were grouped by source IP address and ranked according to the total number of requests. High request counts may indicate automated activity, scanning behaviour, or abnormal network communications.

```

index=botsv3 earliest=0
(sourcetype=*web* OR sourcetype=*access* OR sourcetype=*stream:http*)
| eval src_ip=coalesce(src, src_ip, c_ip, clientip)
| stats count as requests by src_ip
| sort - requests

```

The screenshot shows the Splunk Enterprise search interface. The search bar contains the following query:

```

earliest=0
(sourcetype=*web* OR sourcetype=*access* OR sourcetype=*stream:http*)
| eval src_ip=coalesce(src, src_ip, c_ip, clientip)
| stats count as requests by src_ip
| sort - requests

```

The search results show 31,423 events. The table displays the top 14 source IP addresses by request count.

src_ip	requests
172.16.0.149	4600
172.16.1.239	2616
192.168.3.130	2331
172.16.197.137	841
91.207.175.249	839
192.168.24.128	581
192.168.70.186	512
172.16.133.131	450
172.31.12.76	401
192.168.8.112	385
192.168.8.103	286
192.168.247.131	255
12.196.122.127	237
192.168.56.10	237

The analysis identified 172.16.0.149 as the highest-volume source with 4,600 requests, followed by 172.16.1.239 with 2,616 requests and 192.168.3.130 with 2,331 requests. Although 172.16.197.137 generated a comparatively lower request count (841 requests), it remained a significant investigation lead due to its appearance across multiple log sources and the presence of additional suspicious indicators identified during subsequent analysis.

Analysis of Error-Heavy HTTP Responses

HTTP response codes were analysed to identify clients generating large numbers of failed requests. A high frequency of 403 (Forbidden), 404 (Not Found), or 500 (Internal Server Error) responses often indicates directory brute forcing, web scanning, or attempts to access unavailable or restricted resources.

```
index=botsv3 earliest=0
(sourcetype=*web* OR sourcetype=*access* OR sourcetype=*stream:http*)
| eval src_ip=coalesce(src, src_ip, c_ip, clientip)
| eval http_status=coalesce(status, sc_status, status_code)
| stats count by src_ip http_status
| where http_status=404 OR http_status=403 OR http_status=500
| sort - count
```

The screenshot shows the Splunk Enterprise interface. The search bar contains the following query:

```
index=botsv3 earliest=0
(sourcetype=*web* OR sourcetype=*access* OR sourcetype=*stream:http*)
| eval src_ip=coalesce(src, src_ip, c_ip, clientip)
| eval http_status=coalesce(status, sc_status, status_code)
| stats count by src_ip http_status
| where http_status=404 OR http_status=403 OR http_status=500
| sort - count
```

The search results show 496 events. The table displays the following data:

src_ip	http_status	count
45.7.231.174	404	81
61.75.35.114	404	55
172.16.0.149	404	39
12.196.122.127	404	21
172.16.197.137	404	17
192.168.70.186	404	7
192.168.3.130	404	5
166.216.157.127	404	4
192.168.247.129	404	4
107.77.75.123	404	2
153.197.134.1	404	2

The analysis identified 45.7.231.174 as producing the largest number of HTTP 404 responses (81 events), followed by 61.75.35.114 (55 events) and 172.16.0.149 (39 events). The investigation also identified 172.16.197.137 generating 17 HTTP 404 responses. While this count was lower than other hosts, it contributed to the overall suspicion when considered alongside the host's high number of unique URI requests and activity observed across multiple data sources

Identification of Scanning-Like URI Requests

To identify evidence of web reconnaissance, requests containing administrative or sensitive URI patterns were searched. Common attack targets such as /admin, /login, phpMyAdmin, robots.txt, and .env files were included because attackers frequently probe these locations while searching for vulnerable services.

```
index=botsv3 earliest=0
(sourcetype=*web* OR sourcetype=*access* OR sourcetype=*stream:http*)
| eval src_ip=coalesce(src, src_ip, c_ip, clientip)
| eval uri_path=coalesce(uri, uri_path, url, cs_uri_stem)
| search uri_path="*admin*" OR uri_path="*login*" OR uri_path="*phpmyadmin*" OR uri_path="*.env*" OR
uri_path="*robots.txt*"
| table _time src_ip uri_path status
```

The screenshot shows the Splunk Enterprise interface. At the top, the search bar contains the query: `index=botsv3 earliest=0 (sourcetype=*web* OR sourcetype=*access* OR sourcetype=*stream:http*) | eval src_ip=coalesce(src, src_ip, c_ip, clientip) | eval uri_path=coalesce(uri, uri_path, url, cs_uri_stem) | search uri_path=*admin* OR uri_path=*login* OR uri_path=*phpmyadmin* OR uri_path=*wp-admin* OR uri_path=*.env* OR uri_path=*robots.txt*`. The search results are displayed in a table with columns: `_time`, `src_ip`, `uri_path`, and `status`. The results show several requests targeting administrative and authentication-related resources, including `/admin/images/flags/GB.gif`, `/login.cgi`, and `/wp-content/plugins/svg-vector-icon-plugin/admin/css/wordpress-svg-icon-plugin-style.min.css`.

_time	src_ip	uri_path	status
2018-08-20 15:04:49.282		/admin/images/flags/GB.gif	404
2018-08-20 15:00:59.333	172.16.197.137	/admin/images/flags/GB.gif	404
2018-08-20 15:00:58.116	172.16.197.137	/admin/images/flags/GB.gif	404
2018-08-20 15:00:55.022	172.16.197.137	/admin/images/flags/GB.gif	404
2018-08-20 15:14:43.403		/wp-content/plugins/svg-vector-icon-plugin/admin/css/wordpress-svg-icon-plugin-style.min.css	200
2018-08-20 14:17:17.239	156.220.233.101	/login.cgi	400
2018-08-20 14:34:50.000		/login.cgi	400

The search identified multiple requests targeting administrative and authentication-related resources, including:

- `/admin/images/flags/GB.gif`
- `/login.cgi`
- `/wp-content/plugins/.../admin/...`

These requests are characteristic of reconnaissance behaviour, where an attacker attempts to discover administrative interfaces, login portals, or misconfigured web applications that may be exploited during later stages of an attack.

Analysis of Error-Heavy HTTP Responses

HTTP response codes were analysed to identify clients generating large numbers of failed requests. A high frequency of 403 (Forbidden), 404 (Not Found), or 500 (Internal Server Error) responses often indicates directory brute forcing, web scanning, or attempts to access unavailable or restricted resources.

```
index=botsv3 earliest=0
(sourcetype=*web* OR sourcetype=*access* OR sourcetype=*stream:http*)
| eval src_ip=coalesce(src, src_ip, c_ip, clientip)
| eval http_status=coalesce(status, sc_status, status_code)
| stats count by src_ip http_status
| where http_status=404 OR http_status=403 OR http_status=500
| sort - count
```

The screenshot shows the Splunk Enterprise interface. The search bar contains the following query: `index=botsv3 earliest=0 (sourcetype=*web* OR sourcetype=*access* OR sourcetype=*stream:http*) | eval src_ip=coalesce(src, src_ip, c_ip, clientip) | eval http_status=coalesce(status, sc_status, status_code) | stats count by src_ip http_status | where http_status=404 OR http_status=403 OR http_status=500 | sort - count`. The search results show 496 events. The table displays the following data:

src_ip	http_status	count
45.7.231.174	404	81
61.75.35.114	404	55
172.16.0.149	404	39
12.196.122.127	404	21
172.16.197.137	404	17
192.168.70.186	404	7
192.168.3.130	404	5
166.216.157.127	404	4
192.168.247.129	404	4
107.77.75.123	404	2
153.197.134.1	404	2
157.97.121.69	404	2
172.16.133.131	404	2

The analysis identified 45.7.231.174 as producing the largest number of HTTP 404 responses (81 events), followed by 61.75.35.114 (55 events) and 172.16.0.149 (39 events). The investigation also identified 172.16.197.137 generating 17 HTTP 404 responses. While this count was lower than other hosts, it contributed to the overall suspicion when considered alongside the host's high number of unique URI requests and activity observed across multiple data sources.

A timechart was generated for source IP 172.16.197.137 to examine its web request activity over the investigation period. The visualisation was used to identify bursts of requests that could indicate automated scanning or scripted behaviour. Any sustained increase in request frequency would support the hypothesis that the source was conducting reconnaissance rather than normal user browsing.

```
index=botsv3 earliest=0
(sourcetype=*web* OR sourcetype=*access* OR sourcetype=*stream:http*)
| eval src_ip=coalesce(src, src_ip, c_ip, clientip)
| eval uri_path=coalesce(uri, uri_path, url, cs_uri_stem)
| search src_ip="172.16.197.137"
| sort 0 _time
| table _time sourcetype src_ip dest uri_path status action
| head 15
```

The screenshot shows the Splunk Search interface. At the top, there's a 'New Search' button and a 'Convert to SPL2' button. Below these, a search query is entered in a text box. To the right of the text box, there's a 'Time range: All time' dropdown and a search icon. Below the text box, there's a status bar showing '841 events (before 7/27/26 11:03:32.000 AM)' and 'No Event Sampling'. At the bottom right, there's a 'Smart Mode' dropdown.

```
index=botsv3 earliest=0
(sourcetype=*web* OR sourcetype=*access* OR sourcetype=*stream:http*)
| eval src_ip=coalesce(src, src_ip, c_ip, clientip)
| eval uri_path=coalesce(uri, uri_path, url, cs_uri_stem)
| search src_ip="172.16.197.137"
| sort 0 _time
| table _time sourcetype src_ip dest uri_path status action
| head 15
```

841 events (before 7/27/26 11:03:32.000 AM) No Event Sampling Job

Show: 20 Per Page ▾ Format ▾ Preview: On				
_time ▴	sourcetype ▴	src_ip ▴	dest ▴	uri_path ▴
2018-08-20 09:13:39.388	stream:http	172.16.197.137		
2018-08-20 09:14:04.158	stream:http	172.16.197.137	/	
2018-08-20 09:15:33.252	stream:http	172.16.197.137		
2018-08-20 09:16:34.654	stream:http	172.16.197.137		
2018-08-20 09:45:29.733	stream:http	172.16.197.137		
2018-08-20 09:50:08.251	stream:http	172.16.197.137	/	
2018-08-20 09:50:08.843	stream:http	172.16.197.137	/MFEwTzBNMEswSTAJBgUrDgMCGGUABBBQX6Z6gAidtSefNc6DC00InqPHDQQUD4BhHII	
2018-08-20 09:50:11.750	stream:http	172.16.197.137	/MFEwTzBNMEswSTAJBgUrDgMCGGUABBBQX6Z6gAidtSefNc6DC00InqPHDQQUD4BhHII	
2018-08-20 09:50:11.985	stream:http	172.16.197.137	/GTSGIAG3/MEkWRzBFMEMwQTAJBgUrDgMCGGUABBT27bJyJKBmjX2jXWgnQJKEapsrQ	
2018-08-20 09:50:12.038	stream:http	172.16.197.137	/MFEwTzBNMEswSTAJBgUrDgMCGGUABBS56bKHAoUD%2B0y1%2B0LhPg9JxyQm4gQUf9N	
2018-08-20 09:50:13.203	stream:http	172.16.197.137	/GTSGIAG3/MEkWRzBFMEMwQTAJBgUrDgMCGGUABBT27bJyJKBmjX2jXWgnQJKEapsrQ	
2018-08-20 09:50:13.434	stream:http	172.16.197.137	/MFEwTzBNMEswSTAJBgUrDgMCGGUABBTsWSLjJ8N0Wujis0rUBFV%2Bc%2FAZAQUX2D	
2018-08-20 09:50:13.796	stream:http	172.16.197.137	/MFEwTzBNMEswSTAJBgUrDgMCGGUABBS56bKHAoUD%2B0y1%2B0LhPg9JxyQm4gQUf9N	

S/N	Timestamp	Requested Path
1	2018-08-20 09:12:52.964	/json
2	2018-08-20 09:13:13.283	/
3	2018-08-20 09:13:39.388	/
4	2018-08-20 09:14:04.158	/
5	2018-08-20 09:15:33.252	/
6	2018-08-20 09:16:34.654	/
7	2018-08-20 09:45:29.733	/
8	2018-08-20 09:50:08.251	/
9	2018-08-20 09:50:08.843	/MFEwTzBNMEswSTAJBgUrDgMCGGUABBBQX6Z6gAidtSeFNc6DC00InqPHDQQUD4BhHIIxYdUvKOeNRji0LOHG2eICEAEbr5kbd6HyRDZLbHK6JWk%3D

10	2018-08-20 09:50:11.750	/MFEwTzBNMEswSTAJBgUrDgMCGGUABBBQQX6Z6gAidtSeFNc6DC00InqPHDQQUD4BhHlixYdUvKOeNRji0LOHG2eICEAUJ3IOigZZKuYu5aG6VZQQ%3D
11	2018-08-20 09:50:11.985	/GTSGIAG3/MEkwRzBFME MwQTAJBgUrDgMCGGUABBT27bBjYjKBmjX2jXWGnQJKEapsrQQUD8K4UJpndnaxLcKG0IOgfqZ%2BuksCCGFquezhariY
12	2018-08-20 09:50:12.038	/MFEwTzBNMEswSTAJBgUrDgMCGGUABBS56bKHAoUD%2BOy1%2B0LhPg9JxyQm4gQUf9N1p8Ld7LvwMAnzQzn6Aq8zMTMCEFE%2FuXQ4cLc0QEGNMJMGmf8%3D
13	2018-08-20 09:50:13.203	/GTSGIAG3/MEkwRzBFME MwQTAJBgUrDgMCGGUABBT27bBjYjKBmjX2jXWGnQJKEapsrQQUD8K4UJpndnaxLcKG0IOgfqZ%2BuksCCCZIFdj5jewL
14	2018-08-20 09:50:13.434	/MFEwTzBNMEswSTAJBgUrDgMCGGUABBTWsLjJ8N0Wujis0rUBfVV%2Bc%2FAZAUUX2DPYZBV34RDFIpgKrL1evRDGO8CEAk3oKmu8YG4fOt7G16K1GY%3D
15	2018-08-20 09:50:13.796	/MFEwTzBNMEswSTAJBgUrDgMCGGUABBS56bKHAoUD%2BOy1%2B0LhPg9JxyQm4gQUf9N1p8Ld7LvwMAnzQzn6Aq8zMTMCEGMYDTj7gd4qdA1oxYY%2BEA%3D

First Fifteen Requested Paths

The first fifteen requests made by 172.16.197.137 were extracted and sorted chronologically. Reviewing the request sequence provides context for the source's initial behaviour and helps determine whether it began with ordinary web navigation or targeted administrative resources. The extracted paths should be examined for requests to sensitive directories such as /admin, /login, configuration files, or other application resources that may indicate reconnaissance or vulnerability scanning.

Pivot to DNS, Network, and Endpoint Telemetry

To validate the suspicious web activity identified during Part D, the investigation pivoted to additional telemetry sources using the identified source IP address 172.16.197.137. Rather than limiting the search to source and destination fields, the investigation searched for occurrences of the IP address across the entire dataset to identify related activity recorded in other security logs.

```
index=botsv3 earliest=0
"172.16.197.137"
| stats count by sourcetype
```

The screenshot shows the Splunk Search interface. The search bar contains the query: `index=botsv3 earliest=0 "172.16.197.137" | stats count by sourcetype`. The search results show 47,876 events. The results are displayed in a table with two columns: `sourcetype` and `count`.

sourcetype	count
Script:ListeningPorts	4
XmlWinEventLog:Microsoft-Windows-Sysmon/Operational	5
stream:arp	160
stream:dhcp	38
stream:dns	7008
stream:http	841
stream:icmp	64
stream:igmp	101
stream:ip	12818
stream:smb	26
stream:tcp	5629
stream:udp	5840
symantec:ep:behavior:file	419
symantec:ep:packet:file	2692
symantec:ep:traffic:file	1575
syslog	10656

An "Activate Windows" watermark is visible in the bottom right corner of the screenshot.

The pivot revealed that the suspicious IP address appeared in numerous independent log sources, including network, DNS, and endpoint telemetry. Significant activity was observed in stream:ip, stream:dns, stream:tcp, stream:udp, syslog, and several Symantec Endpoint Protection logs. This demonstrates that the activity associated with the source IP was recorded across multiple security monitoring technologies rather than being limited to HTTP logs.

Hypothesis

Status: Supported

Supporting Observations

1. High Web Activity: The source IP 172.16.197.137 generated 841 HTTP requests and 425 unique URI paths, indicating behaviour consistent with automated web reconnaissance.
2. Cross-Source Network Evidence: The same IP was observed in multiple network telemetry sources, including stream:ip (12,818 events), stream:tcp (5,629 events), stream:udp (5,840 events), stream:dns (7,008 events), and syslog (10,656 events). The presence of the IP across several network logs indicates sustained network activity rather than isolated web requests.
3. Endpoint Evidence: The IP was also present in several endpoint security logs, including symantec:ep:packet:file, symantec:ep:traffic:file, symantec:ep:behavior:file, Microsoft Sysmon Operational, and Script:ListeningPorts, demonstrating that endpoint monitoring systems also observed activity involving the same entity.

The investigation hypothesis is supported. Unlike the initial pivot that relied only on src and dest fields, searching for the IP address throughout the dataset revealed that 172.16.197.137 appears across network, DNS, and endpoint telemetry. Together with its high number of HTTP requests and unusually large number of unique URI requests, the cross-source evidence strengthens the assessment that this host exhibited behaviour consistent with suspicious or potentially malicious activity. Additional examination of the individual DNS, TCP, and endpoint events would be required to determine the precise attack stage and objective.

Cross-Source Incident Reconstruction

The investigation focused on the source IP address 172.16.197.137, which was identified during Part D as the most suspicious entity. This host generated a high volume of unique HTTP requests, accessed numerous URI paths associated with reconnaissance, and appeared in multiple security data sources, including HTTP, DNS, network, endpoint, and Sysmon logs.

Investigation Time Window

The investigation covered activity generated by the selected entity between 20 August 2018 09:3:19 UTC and 20 August 2018 15:18:00 UTC. This time window was selected because it contained the first observed suspicious HTTP request and the subsequent sequence of related network activity identified during the investigation.

```
index=botsv3 earliest=0  
("172.16.197.137" OR src="172.16.197.137" OR dest="172.16.197.137")  
| stats earliest(_time) as First_Event latest(_time) as Last_Event  
| convert ctime(First_Event) ctime(Last_Event)
```

New Search SPL Convert to SPL2 Save As Create Table V

```

index=botsv3 earliest=0
("172.16.197.137" OR src="172.16.197.137" OR dest="172.16.197.137")
| stats earliest(_time) as First_Event latest(_time) as Last_Event
| convert ctime(First_Event) ctime([Last_Event])

```

Time range: All time

✓ 47,876 events (before 7/27/26 12:14:18.000 PM) No Event Sampling Job || [] → [] [] [] []

Events Patterns **Statistics (1)** Visualization

Show: 20 Per Page Format Preview: On

First_Event	Last_Event
08/20/2018 09:03:19	08/20/2018 15:18:00.598

Pivoting Across Log Sources

To reconstruct the incident, the investigation pivoted using the source IP address 172.16.197.137 as the stable investigation entity. The IP address was selected because it was identified as suspicious during Part D. Pivoting enabled related events involving the same entity to be located across multiple log sources, including web, network, DNS, endpoint, and system logs. This approach allowed the sequence of activities associated with the selected entity to be reconstructed chronologically and correlated across different security telemetry.

```

index=botsv3 earliest=0
("172.16.197.137" OR src="172.16.197.137" OR dest="172.16.197.137")
| eval entity=coalesce(src, dest, host, user, ComputerName)
| eval summary=coalesce(signature, action, uri, process_name, EventDescription, _raw)
| sort 0 _time
| table _time sourcetype host entity summary

```

The screenshot shows the Splunk 10.4.1 search interface. The search bar contains the following query:

```
index=botsv3 earliest=0
("172.16.197.137" OR src="172.16.197.137" OR dest="172.16.197.137")
| eval entity=coalesce(src, dest, host, user, ComputerName)
| eval summary=coalesce(signature, action, uri, process_name, EventDescription, _raw)
| sort 0 _time
| table _time sourcetype host entity summary
```

The search results show 47,876 events. The table below displays the first five results, which are identical, indicating a high frequency of duplicate records.

_time	sourcetype	host	entity	summary
2018-08-20 09:03:19	symantec:ep:packet:file	SEPM	SEPM	2018-08-20 09:03:19,PCERF-L,Local: 172.16.197.137,Local: 55811,Remote: 34.215.24.225,Remote: splunk.froth.ly,Remote: 9997,Outbound,Application: C:/Program Files/SplunkUniversalForwarder/bin/splunkd.exe,Action: Allowed
2018-08-20 09:03:19	symantec:ep:packet:file	SEPM	SEPM	2018-08-20 09:03:19,PCERF-L,Local: 172.16.197.137,Local: 55811,Remote: 34.215.24.225,Remote: splunk.froth.ly,Remote: 9997,Outbound,Application: C:/Program Files/SplunkUniversalForwarder/bin/splunkd.exe,Action: Allowed
2018-08-20 09:03:19	symantec:ep:packet:file	SEPM	SEPM	2018-08-20 09:03:19,PCERF-L,Local: 172.16.197.137,Local: 55811,Remote: 34.215.24.225,Remote: splunk.froth.ly,Remote: 9997,Outbound,Application: C:/Program Files/SplunkUniversalForwarder/bin/splunkd.exe,Action: Allowed
2018-08-20 09:03:19	symantec:ep:packet:file	SEPM	SEPM	2018-08-20 09:03:19,PCERF-L,Local: 172.16.197.137,Local: 55811,Remote: 34.215.24.225,Remote: splunk.froth.ly,Remote: 9997,Outbound,Application: C:/Program Files/SplunkUniversalForwarder/bin/splunkd.exe,Action: Allowed
2018-08-20 09:03:19	symantec:ep:packet:file	SEPM	SEPM	2018-08-20 09:03:19,PCERF-L,Local: 172.16.197.137,Local: 55811,Remote: 34.215.24.225,Remote: splunk.froth.ly,Remote: 9997,Outbound,Application: C:/Program Files/SplunkUniversalForwarder/bin/splunkd.exe,Action: Allowed

Field Extraction Using rex

The available Splunk fields, including **src**, **dest**, **uri**, and **status**, provided the required information for the investigation. Therefore, no additional field extraction using the **rex** command was necessary because reliable structured fields were already present within the dataset.

Deduplication

During the incident reconstruction, duplicate records were removed to ensure that each event in the timeline represented a unique activity. A duplicate was defined as multiple log entries that contained the same timestamp, sourcetype, investigation entity (source IP), and event summary (such as the same requested URI or action), indicating that they

represented the same observed activity rather than separate events. Only one representative record was retained for each duplicate event to produce an accurate and concise timeline.

Timeline Classification

Each event identified during the investigation was classified according to its observed behaviour. Requests targeting multiple web resources and administrative locations were classified as Reconnaissance because they indicated attempts to discover accessible resources. Normal DNS queries and certificate validation traffic were classified as Benign, while routine network communications that could not be directly linked to malicious activity were classified as Unknown. Where endpoint security logs recorded network inspection activity, these events were treated as supporting evidence rather than direct proof of attacker actions. ATT&CK mappings were applied only where sufficient evidence existed to justify the classification.

UTC Time	Sourcetype	Host / Entity	Observed Fact	Classification	MITRE ATT&CK
2018-08-20 09:03:19	symantec:ep :packet:file	SEPM	Splunk Universal Forwarder (splunkd.exe) allowed outbound connection from 172.16.197.137 to 34.215.24.225.	Benign	N/A
2018-08-20 09:12:52	stream:http	172.16.197.137	HTTP request made to /json.	Reconnaissance	T1595
2018-08-20 09:13:13	stream:http	172.16.197.137	HTTP request to /.	Reconnaissance	T1595

2018-08-20 09:13:39	stream:http	172.16.197. 137	Additional HTTP request to /.	Reconnaissance	T1595
2018-08-20 09:14:04	stream:http	172.16.197. 137	Continued enumeration of web server.	Reconnaissance	T1595
2018-08-20 09:15:33	stream:http	172.16.197. 137	Repeated HTTP request.	Reconnaissance	T1595
2018-08-20 09:16:34	stream:http	172.16.197. 137	Continued browsing/en umeration activity.	Reconnaissance	T1595
2018-08-20 09:45:29	stream:http	172.16.197. 137	New HTTP session observed.	Reconnaissance	T1595
2018-08-20 09:50:08	stream:http	172.16.197. 137	Long encoded URI requested (OCSP/certif icate validation style request).	Unknown	T1071.001
2018-08-20 09:50:11	stream:http	172.16.197. 137	Additional encoded certificate-re lated URI requested.	Unknown	T1071.001
2018-08-20 09:50:13	stream:http	172.16.197. 137	Multiple encoded HTTP requests generated within seconds.	Reconnaissance / Unknown	T1595
2018-08-20 09:50:14	stream:http	172.16.197. 137	Final encoded HTTP request observed before activity ceased.	Unknown	T1071.001

Data Gaps

Despite successfully reconstructing the incident timeline, several limitations prevented a stronger conclusion from being reached.

The first limitation was the absence of user authentication and identity logs associated with the selected entity. Consequently, it was not possible to determine which user account, if any, initiated the observed network activity.

The second limitation was the lack of detailed endpoint execution evidence directly linking the observed network communications to a specific malicious process. Although endpoint telemetry recorded network activity, it did not provide sufficient evidence to confirm malware execution or persistence on the affected host.

Dashboard and Historical Alert Validation

UTC Time	Sourcetype	Host / Entity	Observed Fact	Classification	MITRE ATT&CK
2018-08-20 09:03:19	symantec:ep :packet:file	SEPM	Splunk Universal Forwarder (splunkd.exe) allowed outbound connection from 172.16.197.137 to 34.215.24.25.	Benign	N/A
2018-08-20 09:12:52	stream:http	172.16.197.137	HTTP request made to /json.	Reconnaissance	T1595
2018-08-20 09:13:13	stream:http	172.16.197.137	HTTP request to /.	Reconnaissance	T1595

2018-08-20 09:13:39	stream:http	172.16.197. 137	Additional HTTP request to /.	Reconnaissance	T1595
2018-08-20 09:14:04	stream:http	172.16.197. 137	Continued enumeration of web server.	Reconnaissance	T1595
2018-08-20 09:15:33	stream:http	172.16.197. 137	Repeated HTTP request.	Reconnaissance	T1595
2018-08-20 09:16:34	stream:http	172.16.197. 137	Continued browsing/en umeration activity.	Reconnaissance	T1595
2018-08-20 09:45:29	stream:http	172.16.197. 137	New HTTP session observed.	Reconnaissance	T1595
2018-08-20 09:50:08	stream:http	172.16.197. 137	Long encoded URI requested (OCSP/certif icate validation style request).	Unknown	T1071.001
2018-08-20 09:50:11	stream:http	172.16.197. 137	Additional encoded certificate-re lated URI requested.	Unknown	T1071.001
2018-08-20 09:50:13	stream:http	172.16.197. 137	Multiple encoded HTTP requests generated within seconds.	Reconnaissance / Unknown	T1595
2018-08-20 09:50:14	stream:http	172.16.197. 137	Final encoded HTTP request observed before activity ceased.	Unknown	T1071.001

Dashboard and Historical Alert Validation

Dashboard

A Splunk dashboard named CSA303_2026/ASA/15643_Investigation was created to visualise the investigation findings. The dashboard contains four panels that summarise the suspicious activity identified during the investigation.

The dashboard includes:

1. **Event Volume** – Displays the number of events over the investigation period.
2. **Top Source Entities** – Shows the source IP addresses generating the highest number of events.
3. **Suspicious HTTP Activity** – Lists suspicious URI requests made by the identified source IP.
4. **Investigation Timeline** – Displays events in chronological order to support incident reconstruction.

The dashboard was configured using historical BOTS v3 data with explicit investigation time boundaries.

Historical Alert Validation

A saved search was created as a detection prototype to identify unusually high HTTP request activity from a single source IP.

Alert Details

Field	Value
Title	Potential HTTP Web Enumeration Detection
Description	Detects source IPs generating unusually high HTTP request volumes that may indicate automated web enumeration or reconnaissance.
Severity	Medium
Owner	Administrator
Schedule	Hourly
Trigger Condition	Number of results greater than zero

Detection Logic

```
index=botsv3 sourcetype=stream:http
| eval path=coalesce(uri, uri_path, url, cs_uri_stem)
| stats count AS Request_Count dc(path) AS Unique_URI_Count BY src_ip
| where Request_Count >= 500
| sort - Request_Count
```

Historical Validation

The saved search was executed against the historical BOTS v3 dataset and successfully identified source IP 172.16.197.137, which generated 752 HTTP requests during the investigation period. This confirmed that the detection logic successfully identified high-volume HTTP activity associated with the investigated entity.

Because the BOTS dataset is historical, the saved search serves as a detection prototype rather than evidence of a real-time alert firing.

Productionisation Note

For deployment within a production Security Operations Centre (SOC), the detection would require the following:

- **Required data sources**
 1. Web server logs
 2. Proxy logs
 3. IDS/IPS logs
 4. Firewall logs
 5. DNS logs
- **Search frequency**
 1. Every 15–60 minutes.
- **Lookback period**
 1. Previous 60 minutes.
- **Expected entities**
 1. Source IP address
 2. Destination server
 3. Requested URI
 4. HTTP status code
 5. User agent

- **Potential false positives**

1. Vulnerability scanners
2. Internal monitoring systems
3. Search engine crawlers
4. Backup or management servers

- **Tuning recommendations**

1. Whitelist approved scanners.
2. Exclude known internal service accounts.
3. Adjust request thresholds based on normal traffic patterns.

- **Response actions**

1. Validate the source IP.
2. Review requested URIs.
3. Pivot to DNS, firewall and endpoint logs.
4. Block or isolate malicious hosts if confirmed.
5. Escalate the incident for further investigation.

The saved search was validated using historical BOTS v3 data and therefore represents a detection prototype. Although the detection logic successfully identified suspicious HTTP activity during historical testing, it does not demonstrate that a scheduled real-time alert fired in a production environment. Continuous monitoring with live data sources would be required before operational deployment.

Findings and Analysis

The investigation identified suspicious HTTP activity originating from source IP 172.16.197.137. Analysis showed unusually high web request activity and a large number of unique URI requests, indicating possible reconnaissance or web enumeration behaviour. Cross-source analysis was performed; however, no additional DNS, IDS or firewall evidence was found to confirm malicious exploitation. Therefore, the investigation concluded that the observed behaviour was suspicious but remained unresolved due to insufficient supporting telemetry.

- Source IP 172.16.197.137 generated 752 HTTP requests during the investigation period.
- The source accessed numerous unique URI paths, indicating possible automated directory enumeration or reconnaissance.
- HTTP logs provided the primary evidence supporting suspicious activity.
- Timeline reconstruction showed repeated reconnaissance-related activity across multiple log sources.
- No DNS, IDS or firewall events were found to independently validate the observed HTTP activity.
- Historical validation confirmed that the developed detection rule successfully identified high-volume HTTP request behaviour.
- The investigation hypothesis remained unresolved because additional evidence was unavailable.

Challenges

Several practical challenges were encountered while investigating Splunk. These challenges affected the investigation process and required additional troubleshooting before meaningful results could be obtained.

- Understanding how Splunk events and fields are related was initially difficult, as clicking on fields such as `c_ip`, `src_ip`, and `dest_ip` returned over 30,000 events, making it challenging to determine whether they were associated with the selected event.
- Identifying the actual suspicious IP address required multiple searches because different sourcetypes stored IP addresses in different field names.
- Building the investigation timeline was challenging due to repeated events that had to be manually reviewed and deduplicated before constructing the final timeline.
- Correlating HTTP activity with DNS, IDS, and network logs was difficult because several searches returned "No results found," making it impossible to fully validate the hypothesis.
- Some required fields were unavailable, requiring manual inspection of `_raw` events and additional field discovery before analysis could continue.
- Creating the Splunk dashboard required several adjustments, particularly when selecting the correct visualisations and modifying SPL searches for each panel.
- Exporting the dashboard as a PDF did not preserve the graphical charts correctly, so dashboard screenshots had to be used instead.
- Developing the saved search (detection prototype) required multiple attempts because the initial SPL searches returned no results until the correct fields and thresholds were identified.

- Understanding some of the assignment requirements, particularly Part E (Cross-Source Incident Reconstruction) and Part F (Dashboard and Detection Validation), required additional interpretation before they could be completed correctly.
- Working with historical BOTS data limited the ability to verify whether the detection would trigger in a real-time production environment.

Recommendations

Future investigations can be improved by increasing log visibility and deploying the developed detection logic within a production monitoring environment.

- Enable collection of firewall, IDS, DNS, proxy and endpoint logs.
- Deploy the HTTP enumeration detection rule as a scheduled production alert.
- Tune request thresholds to reduce false positives.
- Whitelist authorised scanners and monitoring systems.
- Perform regular threat hunting for reconnaissance behaviour.
- Correlate web logs with endpoint and authentication logs during investigations.

Conclusion

The investigation successfully identified suspicious HTTP activity associated with source IP 172.16.197.137. While the observed behaviour was consistent with reconnaissance or web enumeration, insufficient supporting evidence from DNS, IDS and firewall logs prevented confirmation of a successful attack. The developed dashboard, timeline and detection prototype provide a repeatable approach for identifying similar activity in future investigations.